



A rule-driven method for disassembly sequence real-time planning of intelligent mixed-flow disassembly line

Xin Chen^a, Shuo Zhu^{b,*}, Zhigang Jiang^a, Wei Yan^c, Yanan Wang^d, Hua Zhang^e

^a Key Laboratory of Metallurgical Equipment and Control Technology, Ministry of Education, Wuhan University of Science and Technology, Wuhan 430081, China

^b Hubei Key Laboratory of Mechanical Transmission and Manufacturing Engineering, Wuhan University of Science and Technology, Wuhan 430081, China

^c School of Automobile and Traffic Engineering, Wuhan University of Science and Technology, Wuhan 430065, China

^d School of Engineering, Deakin University, Geelong, Victoria 3216, Australia

^e Academy of Green Manufacturing Engineering, Wuhan University of Science & Technology, Wuhan 430081, China

ARTICLE INFO

Keywords:

Mixed-flow disassembly
Disassembly sequence planning
Real-time planning
Disassembly rule (DR)
Improved gene expression programming (IGEP)

ABSTRACT

Intelligent mixed-flow disassembly serves an important role in enhancing the flexible operation capability of the disassembly line for multi-variety and small-batch used electromechanical products, and in forming the scale benefit of recycling. However, in mixed-flow disassembly lines, pre-planning the disassembly sequence for products is typically unfeasible and can lead to “over-disassembly” or “under-disassembly”, resulting in economic losses. This is attributed to significant variations in disassembly features, such as internal structural damage and subassembly recycling income, even within the same type of used electromechanical product. These differences contribute to considerable uncertainties in disassembly sequence planning. To this end, a method is proposed to leverage dynamic disassembly feature information for mining disassembly rules, efficiently driving disassembly sequences real-time planning, and realizing intelligent optimization of product disassembly profit on mixed-flow disassembly line. First, a multi-matrix collaboration based dynamic representing method of product disassembly feature information is proposed. It aims to eliminate the influence of uncertain information on the disassembly sequence planning by recording real-time disassembly feature information, such as disassembly operation time and subassembly recycling income. On this basis, a rule-driven disassembly sequences real-time planning model is built. This model realizes disassembly sequence real-time planning based on offline mined rules using the updated disassembly feature information in matrices. Finally, an improved gene expression programming-based disassembly rule offline mining algorithm is proposed, which incorporates a mutation probability mapping function to improve the efficiency and effectiveness of disassembly rule mining. A case study containing two products with differences in structure and function are used to confirm the effectiveness of the proposed method.

1. Introduction

With the rapid development of industry, the global total volume of used electromechanical products is dramatically increasing at an annual growth rate of 3 %–5%. These used electromechanical products can cause serious environmental pollution if they are not handled properly (Guo, Zhou, Abusorrah, Alsokhry, & Sedraoui, 2020). Meanwhile, as an important “urban mineral”, used electromechanical products contain important resources such as copper, aluminum, iron, plastic, glass, etc., which have extremely high recycling value (Zeng, Liang, & Duan, 2023). In recent years, due to the continued development of industries such as recycled metals, renewable plastics and glass processing, as well as the rising prices of commodities such as copper and aluminum, the market

size of the disassembly and recycling industry of used electromechanical products has been growing rapidly.

In the recycling process of used electromechanical products, disassembly sequence planning is crucial. It directly determines the product recovery rate and reuse rate. By rationally planning the disassembly sequence, the recycling rate of materials can be maximized, the cost of labor and equipment can be reduced, and the environmental impact can be reduced. How to develop and optimize the disassembly sequence has become a hot research issue in the field of recycling (Gong et al., 2019). Presently, the disassembly of used electromechanical products often involves pre-screening, and utilizes a pre-defined disassembly sequence to uniformly treatment products with similar failure forms (Zhou et al., 2019). However, due to the diversity of products and failure forms, the

* Corresponding author.

E-mail address: zhushuo@wust.edu.cn (S. Zhu).

<https://doi.org/10.1016/j.cie.2024.110666>

Received 18 April 2024; Received in revised form 15 October 2024; Accepted 16 October 2024

Available online 24 October 2024

0360-8352/© 2024 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

establishment of an independent disassembly line for each product or even each failure form leads to a serious waste of resources, greatly limiting the scale and efficiency of disassembly. Mixed-flow disassembly line possesses flexible processing capabilities, enabling it to accommodate a variety of types and failure forms of used electromechanical products. Therefore, the mixed-flow disassembly line has become a critical development strategy for enterprises to enhance flexible disassembly capabilities and achieve scale recovery benefits (Ong, Chang, & Nee, 2021).

However, within intelligent mixed-flow disassembly lines, there are uncertain differences in the sources, types, structures, and performance residual values of used electromechanical products (Chand & Ravi, 2023; Mao et al., 2021). This leads to the possibility of both over-disassembly (where costs increase due to unnecessary disassembly operations despite no recycling value) and under-disassembly (where the absence of necessary disassembly operations reduces disassembly profits despite recycling value) in the pre-planned disassembly sequence, making it difficult to guarantee the product disassembly benefit on the mixed-flow disassembly line, and even lead to the loss of recycling economic benefits. Therefore, it becomes critical to make a decision after each disassembly operation to determine whether to continue the disassembly. This type of disassembly is called selective disassembly. Considering the frequent decision-making needs of selective disassembly, real-time planning of disassembly sequences is required to ensure the product disassembly profits on the mixed-flow disassembly line.

As the subassemblies are progressively removed, the internal structural state and degree of wear of the used product are revealed gradually during the disassembly process. The wear degree of disassembles and connecting structures can be utilized to calculate the disassembly operation time and the subassembly recycling income, thereby transforming the uncertain disassembly feature information within the product into definite information. The disassembly sequence real-time planning method can real-time generation disassembly sequence by leveraging this information, thereby eliminating the influence of uncertain information on disassembly sequence planning. However, the effective recording and representation of this acquired disassembly feature information in a dynamic manner present a challenge.

Meanwhile, real-time disassembly sequence planning requires frequent decision-making before each disassembly operation to determine the next disassembly operation, thus imposing stringent requirements on decision-making time. Conventional heuristic-based replanning methods require multiple optimization solutions to continuously adjust the disassembly sequence, resulting in long response times and struggling to meet the real-time demands of frequent disassembly decisions (Nie, Shao, Gao, & Li, 2010). The application of hyper-heuristic methods to mine rules for real-time decision-making is widely adopted in various fields such as job shop scheduling, vehicle routing, and air traffic control to mitigate delays caused by heuristic-based methods (Branke, Nguyen, Pickardt, & Zhang, 2015). Hyper-heuristic methods, unlike heuristic methods, do not seek a specific solution to a given problem but instead aim to find generalized rules, for solving a class of problems. These rules, extracted from historically successful decisions, demonstrate robust reusability and generalization capabilities, enabling them to effectively solve similar unseen problems (Drake, Kheiri, Özcan, & Burke, 2020). They are typically expressed as priority functions that contain various problem domain parameters and symbols. Once generated, these rules are capable of making real-time decisions by calculating the priority of actions. Thus, rule-driven real-time decision processes require only a few milliseconds to execute (Zhao et al., 2023).

This provides a good idea for solving the selective disassembly sequences real-time planning of mixed-flow disassembly line. The speed advantage of rule-based decision-making is used to address the frequent decision-making demands of selective disassembly. By utilizing the reusability and generalization capability of rules to solve the problem of

generating disassembly sequences for used electromechanical products of the same type but with different failure form. However, as far as we know, there are no studies that apply hyper-heuristic method mining disassembly rules (DRs) to solve selective disassembly sequence real-time planning problems.

Therefore, a rule-driven disassembly sequence real-time planning method is proposed for the first time in combination with the characteristics of the disassembly sequence planning problem. This method utilizes offline mined optimal DR to enable real-time disassembly sequences planning in mixed-flow disassembly lines. Specifically, the research problems and objectives of this paper are as follows:

1. Product uncertainty dynamic representation. Design product representation matrices to dynamically represent disassembly feature information. By acquiring and recording information in real-time, matrices eliminate the influence of uncertain information on disassembly sequence real-time planning

2. Selective disassembly sequence real-time planning model construction. Build a rule-driven selective disassembly sequence real-time planning model to real-time generation the disassembly sequence by using dynamically updated disassembly feature information in matrices, to avoid “over-disassembly” and “under-disassembly” and maximize the product disassembly profit

3. Efficient disassembly rule mining algorithm establishment. Propose an improved gene expression programming (IGEP) algorithm to mine optimal DR and enhance the rule mining efficiency and effectiveness of the algorithm

This paper is organized as follows. Section 2 reviews the related work. The rule-driven online planning and offline mining framework in mixed-flow disassembly lines is described in Section 3. Section 4 proposes a multi-matrix collaborative product representation method to record dynamic update disassembly feature information. Section 5 presents a rule-driven disassembly sequences real-time planning model. Section 6 designs an IGEP-based disassembly rules mining algorithm. Section 7 carries out two test cases and analyzes experimental results. Finally, Section 8 concludes this article.

2. Literature review

The relevant studies of used electromechanical product disassembly sequence planning are reviewed from two perspectives concerning: (1) Product disassembly feature information representation, and (2) Disassembly sequence planning method. The challenges faced by using electromechanical products disassembly sequence real-time planning are analyzed and the research gap is identified.

2.1. Product disassembly feature information representation

The computer is unable to directly identify the structure and state of used electromechanical products. Consequently, it is crucial to convert these products into a recognizable form, which representing their true state for program interpretation. Researchers commonly assume prior knowledge of the internal state within the product. Two primary disassembly feature information representation methods, matrix representation and graph representation (such as disassembly trees (Güngör & Gupta, 2001), disassembly network graphs (Yu et al., 2017); AND/OR graphs (Münker & Schmitt, 2022); interference matrix (Mircheski, 2019), have been proposed based on this assumption.

However, the diverse service environment of used electromechanical products introduces significant uncertainty into their internal state. Studies have responded to this challenge by integrating stochastic, fuzzy data, and probabilities into existing matrix and graph representation methods to effectively capture and describe this uncertainty. Notable examples include the work of Guo et al. (Guo et al., 2022), who employed stochastic values to delineate disassembly-related factors such as subassembly recycling income, energy consumption, and carbon emissions. Similarly, Slama et al. (Slama, Ben-Ammar, Dolgui, &

Masmoudi, 2021) utilized discrete random variables with known and bounded probability distributions to model disassembly lead time. Fu et al. (Fu, Zhou, Guo, Qi, & Sedraoui, 2021) adopted failure probability to represent potential disassembly operations that could not be successfully executed during disassembly. Laili et al. (Laili, Ye, Wang, & Zhang, 2021) created a deformation probability matrix to characterize the uncertainty existing in product structure. Behdad et al. (Behdad et al., 2012, 2014) considered uncertain disassembly factors such as time, cost, and the probability of incurring damage when determine the best disassembly sequence for products.

Nevertheless, during the disassembly process, the internal structure and subassemblies of used electromechanical products are gradually exposed, transforming the original uncertainty information to certainty. Therefore, the disassembly process of used electromechanical products is a dynamic process for obtaining internal disassembly feature information. In comparison to representation methods that use stochastic, fuzzy data, and probabilities to describe the internal uncertainty of products, a method that dynamically updates disassembly feature information does not rely on abstract probabilities or fuzzy reasoning, thus enhancing the credibility of planning. How to represent and record the dynamically updated disassembly feature information during the disassembly process is a key research issue.

2.2. Disassembly sequence planning method

Disassembly sequence planning aims to achieve specific objectives, such as minimizing time, reducing costs, maximizing economic value, and meeting environmental indicators (Ong et al., 2021). Various optimization methods, including the Linear Programming Method (LPM) (Ma, Jun, Kim, & Lee, 2011; Ullerich & Buscher, 2013), Genetic Algorithms (GA) (Lee, Tseng, Chang, & Huang, 2020), Particle Swarm Optimization (PSO) (Gulivindala, Bahubalendruni, Varupala, & Ravi, 2021), Ant Colony Optimization (ACO) (Xing, Wu, & Qu, 2021), and Differential Evolution (DE) (Kucukoc, Keskin, & Li, 2021) have been employed to find the optimal disassembly sequence.

However, the actual situation of used electromechanical products is often different from the initial conditions. To ensure product disassembly profit, it is crucial to make timely decisions in response to the certainties found in the dynamic updating of disassembly features information during the disassembly process. Therefore, dynamic adjustment of disassembly sequence is necessary when the internal disassembly feature information of used electromechanical products has changed. Ye et al (Ye, Perrett, Zhang, Laili, & Wang, 2022) proposed an iterative re-planning strategy based dual-loop self-evolving algorithm to address disassembly sequence planning problem under uncertain interference conditions. ElSayed et al. (ElSayed, Kongar, Gupta, & Sobh, 2012) proposed a GA-based dynamically adjusting disassembly sequences re-planning method. Laili et al. (Laili, Tao, Pham, Wang, & Zhang, 2019) introduced a disassembly re-planning method by combining a two-pointer detection strategy and a super-fast bees algorithm to identify detachable elements.

As the complexity of products increases, heuristic-based re-planning methods require frequent adjustments to disassembly sequences, resulting in significant time consumption. For intelligent mixed-flow disassembly lines with diverse product types and failure forms, frequent re-planning is difficult to meet the real-time demand for decision-making, which easily leads to production chaos. Unlike re-planning, real-time planning does not generate the disassembly sequence in advance but makes real-time decisions based on the current disassembly state of the product, selecting the next disassembly operation to gradually build the disassembly sequence. As a result, real-time planning typically completes decisions within a few milliseconds (Zhao et al., 2023). Nonetheless, current real-time planning methods heavily rely on workers to make decisions based on dynamically acquired disassembly feature information and empirical knowledge. This knowledge is often implicit and challenging to formalize and transfer to

others, which significantly limits the development of real-time planning. Therefore, constructing a rule-driven disassembly sequence real-time planning model and mining explicit disassembly rules from a large amount of historical data generated during the disassembly process is a key issue that needs to be urgently addressed.

2.3. Summary of literature gaps

Table 1 and Table 2 summarize the research status on both product disassembly feature information representation and disassembly sequence planning methods. While many scholars acknowledge the importance of disassembly sequences real-time planning, there remain several research gaps requiring attention. These gaps primarily manifest in two aspects: product representation and planning method. Regarding product representation, there is a deficiency in effective methods for representing disassembly features, specifically in recording dynamic updates of disassembly operation time and subassembly recycling income during the disassembly process. On the planning method front, a scarcity exists in methods that can utilize dynamically acquired disassembly feature information to realize disassembly sequence real-time planning. Specifically, the contributions of this work are outlined as follows.

(1) To record the internal disassembly feature information obtained dynamically during the disassembly process, multi-matrix collaborative based disassembly feature information representation matrices are designed. There are three types of matrices: activation matrix, association matrix and disassembly feature matrix. These matrices record the disassembly feature information acquired in real time to eliminate the influence of uncertain information on the disassembly sequence planning. Importantly, this method eschews abstract probabilities or fuzzy reasoning, thereby bolstering the reliability of the planning process.

(2) To realize disassembly sequence real-time planning, a rule-driven disassembly sequence real-time planning model is built for the first time. This model utilizes dynamically updated disassembly feature information to achieve real-time generation of the disassembly sequence through real-time decision making. After completion previous disassembly operation and faced with multiple subsequent executable disassembly operations, the DR calculates the priority of each disassembly operation (including the termination operation) to determine the next disassembly operation. The loop terminates when the termination operation is selected or no disassembly operation can be performed. This decision-making process helps in determining whether to continue the disassembly and which specific disassembly operation to

Table 1

Summary of selected existing research on product disassembly feature information representation.

Author(s)	Representation method	Uncertainty representation
Güngör and Gupta (Güngör & Gupta, 2001)	Disassembly tree	—
Yu et al. (Yu et al., 2017)	Disassembly network graph	—
Münker and Schmitt (Münker & Schmitt, 2022)	AND/OR graph	—
Mircheski (Mircheski, 2019)	Interference matrix	—
Guo et al. (Guo et al., 2022) and Slama et al. (Slama et al., 2021)	AND/OR graph	Random variables with known and bounded probability distributions
Fu et al. (Fu et al., 2021)	AND/OR graph	Failure probability
Laili et al. (Laili et al., 2021)	Interference matrix	Deformation probability
Behdad et al. (Behdad et al., 2012, 2014)	AND/OR graph	Probability
Current research	Multi-matrix collaborative	Dynamic update information

Table 2

Summary of selected existing research on disassembly sequence planning method.

Author(s)	Planning type	optimization method
Ullerich and Buscher (Ullerich & Buscher, 2013) and Ma et al. (Ma et al., 2011)	Static planning	LPM
Lee et al. (Lee et al., 2020)	Static planning	GA
Gulivindala et al. (Gulivindala et al., 2021)	Static planning	PSO
Xing et al. (Xing et al., 2021)	Static planning	ACO
Kucukkoc et al. (Kucukkoc et al., 2021)	Static planning	DE
Ye et al. (Ye et al., 2022)	Re-planning	Dual-loop self-evolving algorithm
ElSayed et al. (ElSayed et al., 2012)	Re-planning	GA
Laili et al. (Laili et al., 2019)	Re-planning	Super-fast bees algorithm
Current research	Real-time planning	Hyper-heuristic algorithm (GEP)

execute.

(3) To obtain the optimal disassembly rule that drive disassembly sequence real-time planning model, an IGEP-based disassembly rule mining algorithm is proposed. Specifically, a mutation probability mapping function is utilized to enhance the mutation operator by assigning varying mutation probabilities to disassembly rules based on their fitness. This approach assists low-fitness rules in breaking through local optima and safeguards the gene combinations of high-fitness rules, thereby enhancing the efficiency and effectiveness of rule mining.

3. Mixed-flow disassembly line disassembly sequence real-time planning

In this section, the problem of disassembly sequences real-time planning on a mixed-flow disassembly line is described. Next, a rule-driven online planning and offline mining framework for disassembly sequence planning of mixed-flow disassembly line is given. Additionally, detailed discussions on the online planning process and the offline mining process are provided.

3.1. Disassembly sequence real-time planning problem description

In the mixed-flow disassembly line, various types of used electromechanical products awaiting disassembly. These products have varying degrees of internal wear, leading to differences in subassembly recycling income and disassembly operation time. Each product has multiple available disassembly sequences. This article aims to maximize the disassembly profit of individual products by reasonable planning their disassembly sequences, thereby achieving maximum the total disassembly profit of mixed-flow disassembly line. In this paper, we make the following assumptions: 1. Sequential and selective disassembly are considered; 2. The influence of connection structure and subassembly wear degree on the disassembly time and recovery income is considered, and their identification methods are realized in advance as a priori knowledge; 3. The setup time between different disassembly operations is considered. To further describe the investigated problem, some notations are listed as follows:

(1) Indices:

s	Index of subassemblies, $s = 1, 2, \dots, S$;
j, k	Index of operations, $j, k = 0, 1, 2, \dots, J$, where 0 is a dummy operation;
m, n	Index of the sequence number of operations in the disassembly sequence, $m, n = 1, 2, \dots, J$.

(2) Notations:

S	Total number of subassemblies;
J	Total number of operations;
O_j	Disassembly operation j ;
O_{stop}	Terminating operation;
A_k	Immediate predecessor operation set for operation k ;
B_K	Mutually exclusive operation set for operation k ;
C	Cost per unit time;
R_s	Recycling/reuse value of subassembly s ;
T_j	Disassembly operation time when performing operation j ;
t_{jk}	Setup time of performing operation k if it is performed after operation j ;
A	Activation matrix of a given AND/OR graph;
I	Incidence matrix of a given AND/OR graph;
I_{sj}	Element in the s th row and j th column of I .

(3) Decision variables:

x_j	If operation j is performed $x_j = 1$; otherwise $x_j = 0$;
y_{jm}	If operation j is the m th operation in the disassembly sequence, $y_{jm} = 1$; otherwise $y_{jm} = 0$.

A disassembly sequence real-time planning problem with the aim of maximum product disassembly profit can be established as follows:

$$\max f = \sum_{j=1}^J \sum_{s=1}^S I_{sj} R_s x_j - C \sum_{j=1}^J T_j x_j - C \sum_{j=1}^J \sum_{k=1}^J \sum_{m=1}^{J-1} t_{jk} y_{jm} y_{k,m+1} \quad (1)$$

$$\text{s.t. } x_0 = 1 \quad (2)$$

$$x_j + x_k \leq 1, j \in B_k, k = 0, 1, 2, \dots, J \quad (3)$$

$$\sum_{j \in A_k} \sum_{m=1}^{n-1} y_{jm} \geq y_{kn}, k = 1, 2, \dots, J \quad (4)$$

$$x_j = \sum_{m=1}^J y_{jm}, j = 1, 2, \dots, J \quad (5)$$

$$\sum_{j=1}^J y_{jm} \leq 1, m = 1, 2, \dots, J \quad (6)$$

where Eq. (1) represents that the objective is to maximize product disassembly profit. In Eq. (1), the first item represents the income of recycling the subassembly, and the second and third items are the cost of completing the disassembly operation and the conversion cost between operation. Eq. (2) requires that the dummy operation must be performed because the dummy operation indicates the generation of the product. Eq. (3) prevents mutually exclusive operations from appearing in a disassembly sequence. Eq. (4) guarantees the priority relationship, i.e., at least one immediate predecessor operation needs to be completed before the disassembly operation can be performed. Eq. (5) represents the association between two decision variables. Eq. (6) requires that only one disassembly operation can be performed at any time during the disassembly process, parallel operations are not allowed.

3.2. A rule-driven disassembly sequence real-time planning framework

The mixed-flow disassembly line consists of a series of robots with a visual recognition system. Various types of used electromechanical products with different failure modes are processed sequentially on the mixed-flow disassembly line. Fig. 1 gives a rule-driven online planning and offline mining framework for disassembly sequence planning of mixed-flow disassembly line. This framework consists of two primary components: offline mining and online planning. The offline part focuses on mining optimal disassembly rule from a large dataset of historical data, while the online part utilizes this rule for real-time disassembly sequence planning. These components are detailed as follows:

(1) In the online planning part, four collaboratively updated product representation matrices are constructed to dynamically record

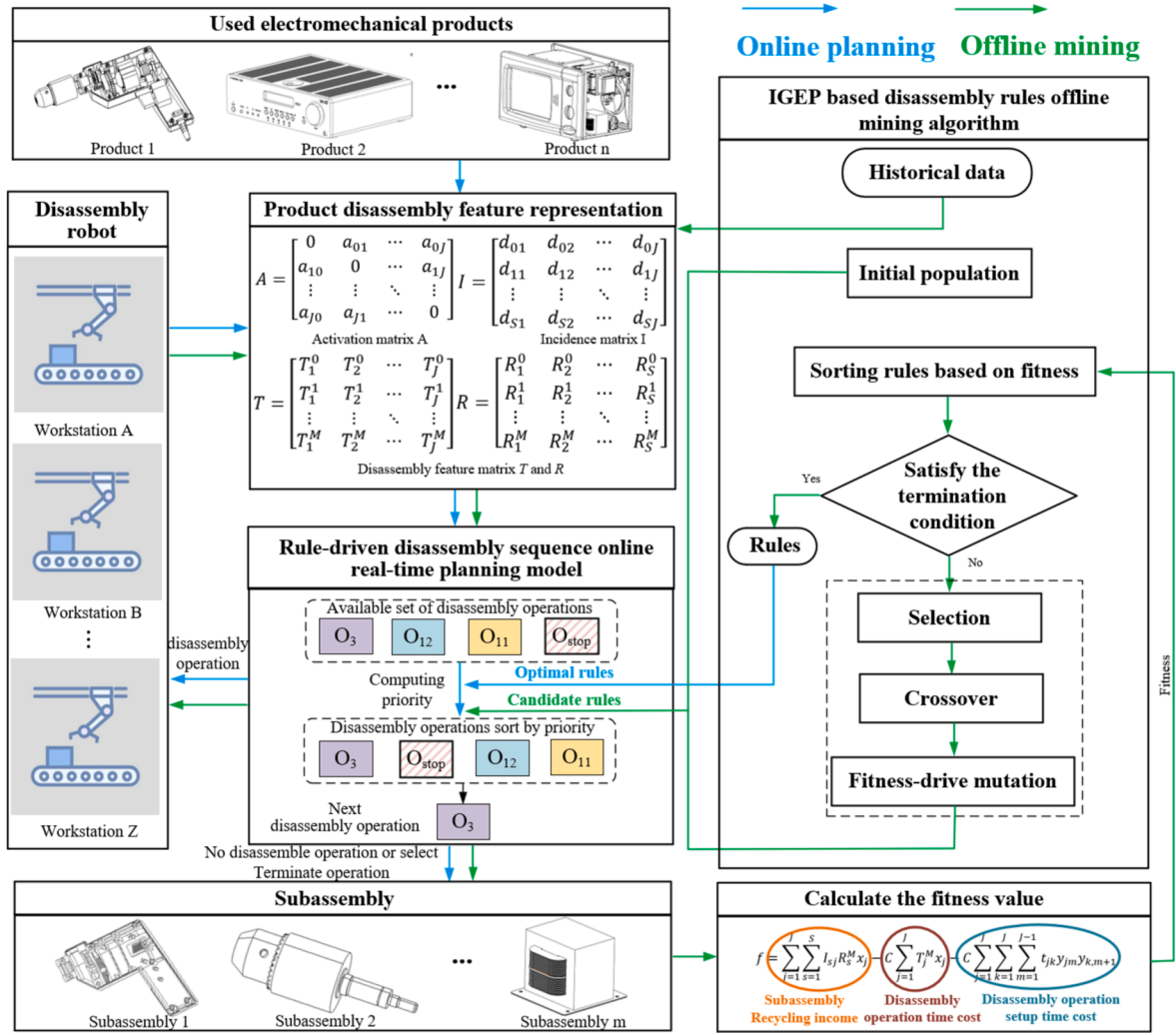


Fig. 1. A rule-driven online planning and offline mining framework for disassembly sequence planning of mixed-flow disassembly lines.

disassembly feature information, such as operation time, recycling income, and constraints related to disassembly operations and subassemblies. Then, the rule-driven real-time disassembly sequence planning model utilizes the information in matrixes to determine the currently executable disassembly operation set, and employs the optimal disassembly rule mined from the offline part to compute the priority of each disassembly operation within this set. The disassembly robot will execute the disassembly operation with the highest priority as the next step. Subsequently, product representation matrices updating and is re-entered into the rule-driven disassembly sequence real-time planning model to determine the subsequent disassembly operation. This iterative process continues until either all subassemblies have been disassembled or a termination operation is selected.

(2) In the offline mining part, historical data will be transformed into product representation matrices. Subsequently, the IGEP algorithm randomly generates an initial population containing multiple disassembly rules. The rule-drive disassembly sequence real-time planning model then converts these rules into the corresponding disassembly sequence based on the data stored in product representation matrices. The product disassembly profit corresponding to these disassembly sequences is calculated and serves as the fitness value for each rule. At this point, the algorithm determines whether the termination condition has been met. If the termination condition has not been met, the disassembly rule population will undergo a series of genetic operations (including selection, crossover, mutation) to generate a new population. The new population will again be input into the rule-drive disassembly sequence

real-time planning model to evaluate the fitness of new rules. Upon satisfaction of the algorithm termination condition, the optimal disassembly rule with the highest fitness is output.

4. Product disassembly feature representation

To eliminate the influence of uncertain information on disassembly sequence planning, it is necessary to record the disassembly feature information that is dynamically updated with the disassembly process. The disassembly features information includes inference relationships between subassemblies, priority conditions, and other information. Therefore, a multi-matrix collaborative based dynamic representation method of product disassembly feature information is designed. Specifically, it constructs activation matrix, incidence matrix, and disassembly feature matrix. The activation matrix and the incidence matrix record the relationships between disassembly operations and the relationships between disassembly operations and subassemblies, respectively. Both matrices are predefined based on the structural constraints of internal subassemblies and remain unchanged during disassembly. In contrast, the disassembly feature matrix records the operation time of disassembly operations and recovery income of subassemblies, and is dynamically updated after each disassembly operation based on the structural disassembly difficulty and the subassembly wear level as identified by the visual recognition system. Three matrices are described as follows:

1) Activation matrix A

The activation matrix A stores the priority and mutual exclusion relationships that exist between disassembly operations. The subsequent disassembly operations that can be selected after each disassembly operation can be determined by analyzing the activation matrix. Specifically, an activation matrix A can be written as follows.

$$A = \begin{bmatrix} 0 & a_{01} & \cdots & a_{0J} \\ a_{10} & 0 & \cdots & a_{1J} \\ \vdots & \vdots & \ddots & \vdots \\ a_{J0} & a_{J1} & \cdots & 0 \end{bmatrix} \quad (7)$$

The columns and rows of the activation matrix represent the disassembly operation. There are three types of relationships among the operations in the activation matrix, and the value of A in these relationships is represented by Eq. (8). It is important to note that the operation 0 in the first row and first column of the matrix represents the initial operation, which is a virtual and startup operation.

$$A_{jk} = \begin{cases} 1, & \text{if operation } k \text{ can be performed after operation } j \\ -1, & \text{if operation } j \text{ and } k \text{ conflict with each other} \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

2) Incidence matrix I

The incidence matrix I stores the correspondence between the disassembly operation and the subassembly. By analyzing the incidence matrix, we can determine which disassembly operations are applicable to subassemblies and the new subassemblies generated after performing these operations. An incidence matrix can be written as follows.

$$I = \begin{bmatrix} d_{01} & d_{02} & \cdots & d_{0J} \\ d_{11} & d_{12} & \cdots & d_{1J} \\ \vdots & \vdots & \ddots & \vdots \\ d_{S1} & d_{S2} & \cdots & d_{SJ} \end{bmatrix} \quad (9)$$

The rows of the incidence matrix represent various subassemblies, while the columns represent different disassembly operations. The I_{sj} in the incidence matrix represents the relationship between operations and subassemblies. The value of I_{sj} is shown in Eq. (10).

$$I_{sj} = \begin{cases} 1, & \text{if subassembly } s \text{ is obtained by operation } j \\ -1, & \text{if subassembly } s \text{ is disassembled via operation } j \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

It is worth noting that the value of I_{sj} in the matrix I is predetermined. Thus, for each particular disassembly operation, the resulting combination of subassemblies is fixed and does not change as the disassembly process proceeds.

3) Disassembly feature matrix T and R

The formula for calculating subassembly recycling income and disassembly operation time based on the degree of wear is as follows:

$$T_j^m = T_j \times (1 + D_j^m) \quad (11)$$

$$R_s^m = R_s \times (1 - W_s^m) \quad (12)$$

In Eq. (11) and Eq. (12) T_j^m and R_s^m represent the disassembly operation time for operation j and the recycling income for subassembly s after the m th disassembly operation. T_j and R_i denote the disassembly operation time required for operation j and the recycling income for subassembly s under the assumption of no wear. D_j^m signify the disassembly difficulty of connection structures after wear. The range of disassembly difficulty is from 0 to 1, where 0 corresponds to minimal disassembly difficulty and 1 corresponds to maximum disassembly difficulty. Additionally, W_s^m signify the wear degree for subassemblies. As the wear degree of the subassemblies W_s^m increases, the subassemblies recycling income decreases.

The dynamic update disassembly feature matrix consists of two parts: the disassembly operation time matrix T and the recycling income matrix R . These two matrices are used to record the disassembly time of each disassembly operation T_j^m and the recycling income R_s^m of each

subassembly, which are dynamically updated after each disassembly operation.

In the matrix T , the rows represent the disassembly operations and the columns represent the number of completed disassembly operations. After each completed disassembly operation, a new row is inserted into matrix T to record the update operation time for disassembly operation. Meanwhile, in matrix R , the rows represent subassemblies and the columns also represent the number of completed disassembly operations. After each disassembly operation, a new row is inserted into matrix R to record the updated subassembly recycling income. It is worth noting that the completed disassembly operations and the corresponding subassembly recycling incomes remain constant in the matrices T and R . Only the disassembly operation time for pending operations and the unobtained subassembly recycling incomes change.

The disassembly feature matrix T and R can be represented by Eq. (13). By recording dynamically acquired disassembly features (disassembly operation times and subassembly recycling incomes), matrices can eliminate the influence of uncertain information on the disassembly sequence planning. This allows for the consideration of dynamically varying operation times and recycling incomes in the optimization process, thus generating disassembly sequences with higher product disassembly profit.

$$T = \begin{bmatrix} T_1^0 & T_2^0 & \cdots & T_J^0 \\ T_1^1 & T_2^1 & \cdots & T_J^1 \\ \vdots & \vdots & \ddots & \vdots \\ T_1^M & T_2^M & \cdots & T_J^M \end{bmatrix} R = \begin{bmatrix} R_1^0 & R_2^0 & \cdots & R_S^0 \\ R_1^1 & R_2^1 & \cdots & R_S^1 \\ \vdots & \vdots & \ddots & \vdots \\ R_1^M & R_2^M & \cdots & R_S^M \end{bmatrix} \quad (13)$$

5. Rule-driven disassembly sequence real-time planning model

To achieve selective disassembly sequences real-time planning, a rule-driven real-time planning model is built. A DR, is essentially a priority function composed of parameters and symbols. In rule-driven disassembly sequences real-time planning model, DR calculates the priority of each disassembly operation based on the parameter values with different disassembly operations and the current state of disassembly. Subsequently, the model selects the operation with the highest priority to execute as the next step. The model gradually constructs the final disassembly sequence by selecting one disassembly operation at a time. Section 6 will provide detailed insight into the constituent elements and symbols of the DRs, along with the method for obtaining them. This section mainly describes the processes to use the rule-drive model to generate disassembly sequence in the disassembly process. The flowchart illustrating the rule-driven disassembly sequence real-time planning model is presented in Fig. 2, with Algorithm 1 providing the corresponding pseudo-code.

Algorithm 1: Pseudo-code of rule-driven disassembly sequence real-time planning model

Input: Disassembly rule (DR); Activation matrix (A); Incidence matrix (I); Disassembly feature matrix (T and R).

Output: Disassembly sequence.

- 1: Initialize the stop-remove variable $ST = 0$ and the number of disassembly operations done $NOD = 0$;
- 2: Determine the set of disassembly operations $Set D(O_i)$;
- 3: **while** ($ST \neq 0$) **do**
- 4: Calculate priority values for all the operations in $Set D(O_i)$ with the help of a DR;
- 5: Select the disassembling operation with the highest priority value;
- 6: **if** highest priority value is termination operation O_{stop} **then**
- 7: **return** final disassembly sequence
- 8: **else**
- 9: Disassembly sequence. append (highest priority operation), $NOD = NOD + 1$;
- 10: Update Activation matrix (A); Incidence matrix (I); Disassembly feature matrix (T and R);
- 11: **end if**
- 12: **end while**
- 13: **return** final disassembly sequence

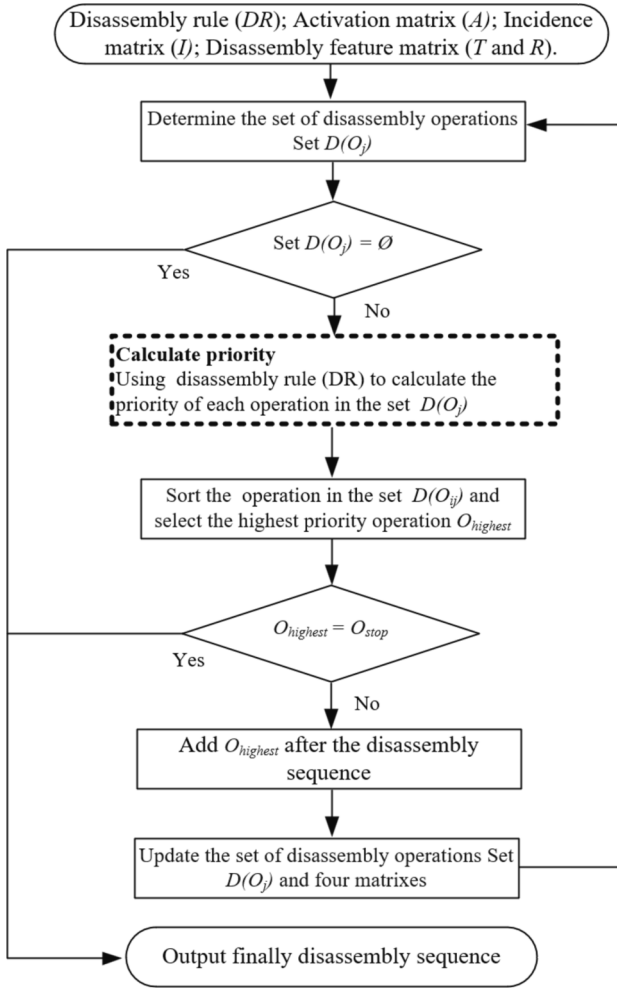


Fig. 2. Rules driven disassembly sequence real-time planning model flow chart.

The specific steps are described as follows:

- (1) The offline mined optimal disassembly rule along with the matrix (A , I , T , and R) are provided as input;
- (2) Determine the Set $D(O_j)$ comprising currently executable disassembly operations;
- (3) If the Set $D(O_j)$ is empty, proceed to Step 9; otherwise, continue to Step 4;
- (4) Calculate the priority of each operation in Set $D(O_j)$ using the offline mined optimal disassembly rule;
- (5) Sort the disassembly operations in Set $D(O_j)$ based on their priority;
- (6) If the highest priority disassembly operation is the termination operation, proceed to Step 9; otherwise, move to Step 7;
- (7) Add the highest priority disassembly operation to the disassembly sequence;
- (8) Update the matrix (A , I , T , and R) with the newly obtained subassembly recycling incomes, disassembly operation times, and other relevant information, and then return to Step 2;
- (9) Output the final disassembly sequence.

Throughout the process, four crucial points need highlight: (a) Each Set $D(O_j)$ contains a termination operation. The termination operation is a virtual operation, and its disassembly operation time, disassembly operation cost, and disassembly income are all 0. (b) The optimal disassembly rule used to calculate the priority of each disassembly operation in Step 4 are mined in advance from historical data. The

specific composition and mining process of these rules will be detailed in Section 6. (c) In Step 8, the four cooperative matrices are updated based on newly disassembly feature information obtained via robot visual recognition, enabling the disassembly rules to real-time decision the subsequent disassembly operation. (d) The entire process comprises two exit mechanisms: Step 3 and Step 6. In Step 6, if the termination operation is assigned the highest priority, it indicates that the DR has concluded that additional disassembly operations would not increase product disassembly profit. Consequently, the disassembly process terminates immediately which is the key for enabling rule-driven real-time planning models to achieve selective disassembly real-time planning.

As an illustration, the Bourjault ballpoint pen (Tian et al., 2018) is employed. Fig. 3 depicts the process of real-time generating disassembly sequences using a rule-driven real-time disassembly sequence planning model. Model undergoes three iterations to get the final disassembly sequence. Initially, when NOD (Number of Operations Done) equals 0, the Set $D(O_j)$ comprises three executable operations: O_1 , O_2 , and O_{stop} . The DR calculates priorities for these three operations, ultimately selects O_1 as the next disassembly operation. Subsequently, the executable disassembly operations within the Set $D(O_j)$ are updated to O_3 , O_{12} , and O_{stop} as NOD is incremented to 1. When NOD reaches 2, the termination operation O_{stop} is selected as the next disassembly operation, resulting in the terminate of the disassembly process. Thus, the final disassembly sequence generated by the model is $O_1-O_{12}-O_{stop}$.

6. IGEP-based disassembly rules mining algorithm

To obtain the optimal disassembly rule to drive the disassembly sequence real-time planning model, an IGEP-based disassembly rule mining algorithmic framework is proposed, comprising a learning module and an evaluation module. Following this, a detailed description of the key components of both the learning and evaluation modules is provided.

6.1. IGEP-based disassembly rules mining algorithm framework

The core aim of the IGEP-based disassembly rules mining algorithm is to mining optimal disassembly rule from a large amount of historical data. The optimal rule can drive rule-based disassembly sequences real-time planning model, thus real-time generation disassembly sequence for the unseen problem. The algorithm contains two modules, the learning module and the evaluation module. Within the learning module, IGEP is employed to drive the evolution of DRs. The evaluation module acts as a performance evaluator, converting the newly generated DRs from the learning module into feasible disassembly sequences and assessing their fitness. Fig. 4 illustrates the framework of the IGEP-based rule mining algorithm.

6.2. Learning module

The learning module consists of three parts, i.e., the selection of disassembly rule elements, the design of coding schemes, and the improvement of genetic operators.

(1) Define disassembly rule element sets

Disassembly rule is a priority function consisting of elements in the function set (FS) and terminal set (TS) relevant to the disassembly sequence planning domain. The elements in the FS and TS are pre-defined. Throughout the evolution process, the algorithm continually adjusts the elements and structure of DR to address the problem at hand. Therefore, the elements in the TS and FS directly affect the quality of the DR. The TS and FS elements used to build DR are shown in Table 3. Terminal elements in TS fall into two categories. One of them is the attribute value of the disassembly operation itself, such as disassembly operation time (DT), setup time (ST), cost of performing the disassembly operation (COP), and recycling income of performing the disassembly (ROP). The other is the current disassembly status value, such as the

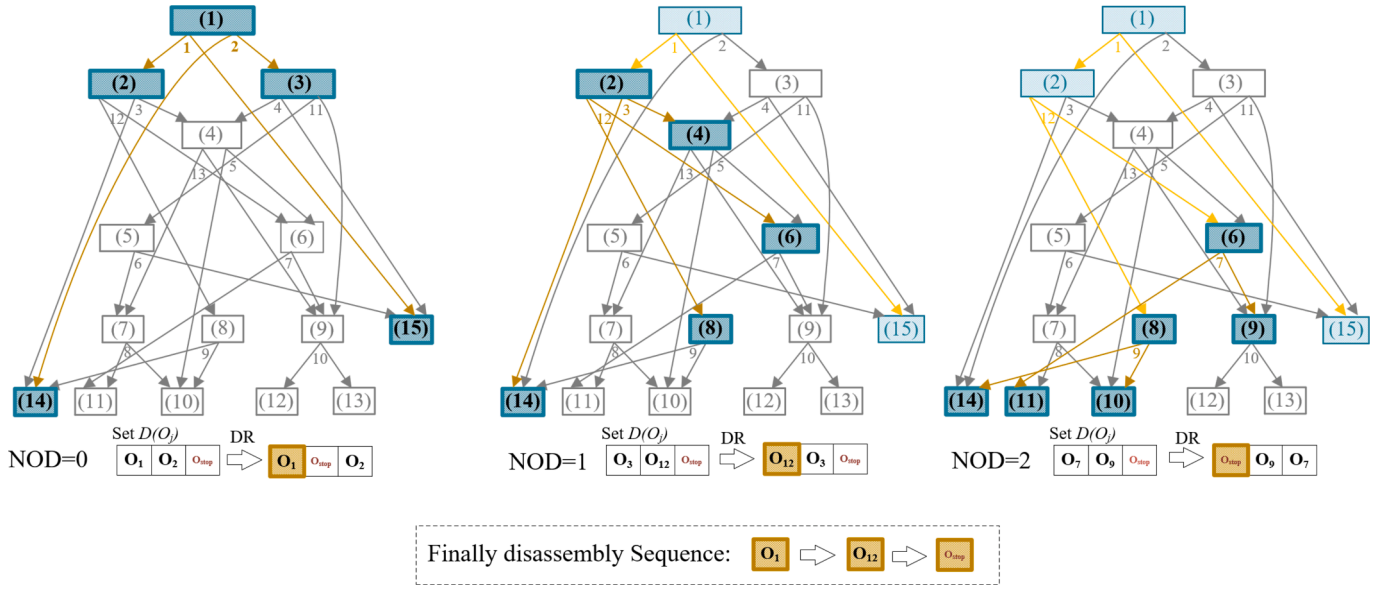


Fig. 3. The process of generating disassembly sequences using rule-driven disassembly sequence real-time planning model.

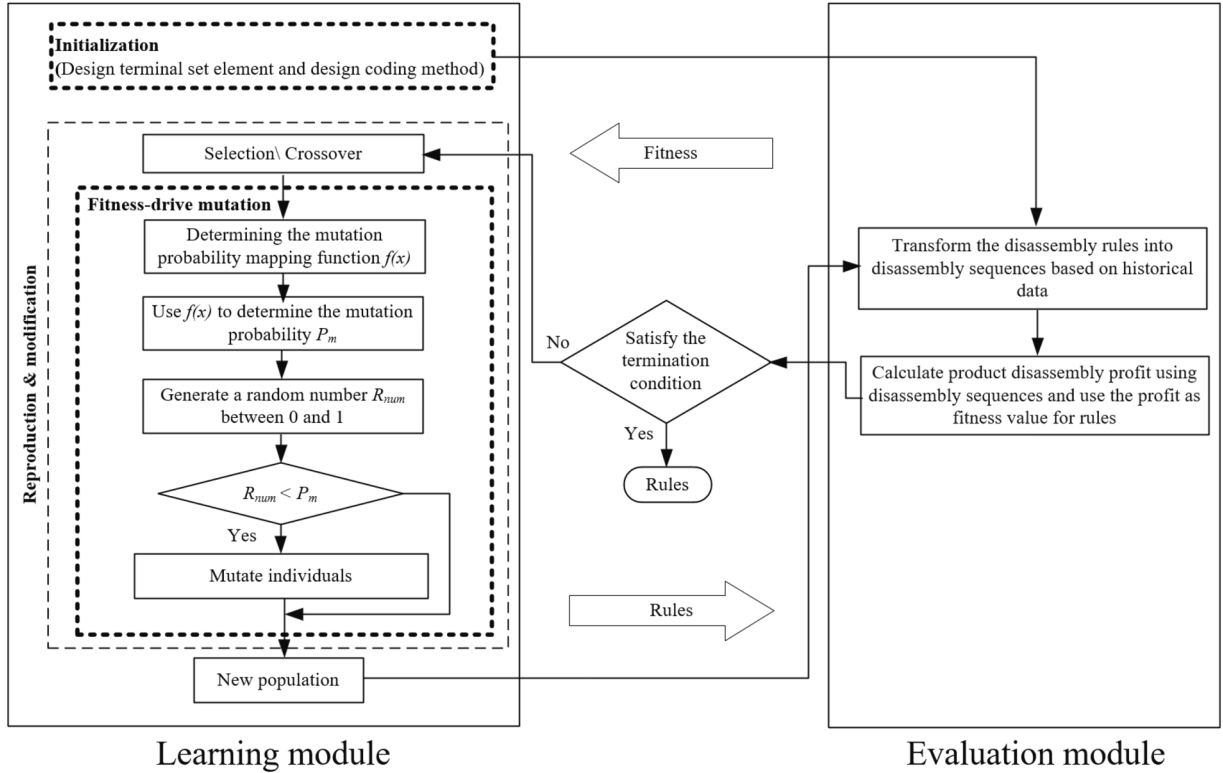


Fig. 4. The framework of the IGEP-based rule mining algorithm.

number of disassembly that have been performed (NOD), current disassembly profit (CDP), the current total cost (CTC), etc. By using these two types of elements, DR can more comprehensively evaluate the priority of each disassembly operation in the current disassembly state, to achieve real-time planning.

The cost of operating is calculated as shown in Eq. (14):

$$COP = C \times (DT + ST) \quad (14)$$

where C is the cost per unit of time. The $/$ is a protective division that returns 1 when the denominator is 0. In addition, DT , ST , COP , and ROP

of the terminating operation O_{stop} are all 0.

(2) Coding schemes

A DR is a priority function that can be encoded into a gene of gene expression programming (GEP). A gene in GEP is a fixed-length symbolic string with a head and a tail. Suppose the gene has h symbols in the head, and t symbols in the tail, then the length of the tail is determined by the equation $t = h * (n - 1) + 1$. The n is the maximum number of arguments for all the operations in FS. Since four functions $+$, $-$, $*$, $/$ are used in FS the value of n is 2. To ensure that each gene can be decoded as a legitimate DR, the head symbolic of a gene can be selected from the FS and

Table 3

Definitions of the functions and terminals element that are used to build DR.

Set	Elements	Meaning
FS	+, −, * /	Express the corresponding arithmetic functions, respectively Expresses the protected division which returns 1 when the denominator is equal to 0
TS	DT ST	Disassembly operation time ($DT_{stop} = 0$) Setup time from the previous operation to the current operation ($ST_{stop} = 0$)
	ADP	The disassembly profit after the completion of the operation
	COP	The cost of performing the disassembly operating ($COP_{stop} = 0$)
	ROP	The recycling income of operating ($ROP_{stop} = 0$)
	NOD	Number of disassembly operations done
	CDP	Current disassembly profit
	CTC	Current total disassembly cost
	CT	Current Time

TS, while the tail symbolic of a gene can only be selected from the TS.

The mapping between gene and DR in GEP is given in Fig. 5. The encoding method of DR contains three states (a) gene, (b) expression tree, and (c) mathematical formula. The DR decodes each symbol in the gene in a left-to-right manner, the first symbol of the chromosome is used as the root node of the expression tree, and the symbol is populated according to the number of branches of the symbol until all the branches are filled. Then the expression tree is converted to a mathematical expression to obtain the final DR. In the figure *gene 1* and *gene 2* have the same length but their final decoded DR are not the same length. This is due to the unique decoding method of the expression tree so that the same length of the gene can be decoded out of different lengths of the DR so that the algorithm has a more flexible solution space to find the optimal DR.

Notably, traditional heuristic algorithms typically encode disassembly sequences directly. However, disassembly sequences must adhere to complex constraints, including operation priority and mutual exclusivity. As a result, heuristic algorithms often produce illegal sequences during initialization, requiring multiple additional steps to adjust and modify the encoding to meet these constraints (Lu et al., 2020). This adjustment process not only increases computational complexity but also introduces additional challenges in maintaining the legality of the code. In contrast, the hyper-heuristic algorithm GEP encodes disassembly rules to generate disassembly sequences. After

randomly generating the initial encoding, any rules corresponding to the code can generate a legal disassembly sequence after passing the rule-driven disassembly sequence planning model. Therefore, the GEP greatly reduces the computational cost of checking and repairing code legality.

(3) Genetic operators

To improve the efficiency of mining disassembly rule in GEP, a fitness-driven mutation operator is designed. The operator uses a mutation probability mapping function to calculate the mutation probability corresponding to different DR fitness. The mutation probability is dynamically adjusted by DR fitness values to make DR with low fitness more prone to mutation, and to increase the ability to explore solution space. DRs with higher fitness have a lower mutation probability, which helps to maintain their excellent genes and reduces the risk of reducing the algorithm convergence speed that may be caused by unnecessary mutations. The fitness-driven mutation operator pseudo-code is shown in algorithm 2.

Algorithm 2: Pseudo-code of fitness-drive mutation

Input: Disassembly rule (DR); Fitness of disassembly rule (F); Mutation probability mapping function ($f(x)$).

Output: New disassembly rule (DR-new).

```

1: Normalize fitness  $F_{new} = (F - F_{min}) / (F_{max} - F_{min})$ ;
2: for  $i$  in population do
3:   Use  $f(x)$  to determine the mutation probability ( $P_m$ ) of DR;
4:   Generate a random number ( $R_{num}$ ) between 0 and 1 for DR;
5:   if  $R_{num} < P_m$  then
6:     DR-new = mutate(population[i]);
7:     Population[i] = DR-new;
8:   end if
9: end for

```

Three mutation probability mapping functions are chosen as shown in Fig. 6. The three mapping functions possess different minimum mutation probabilities (0.05 and 0.01) and maximum mutation probabilities (0.05, 0.1, and 0.2). The exponential function is employed to express the mapping of normalized fitness to mutation probabilities. Because the exponential function amplifies differences in fitness, rules with lower fitness are more likely to be selected for mutation. This helps to accelerate the evolutionary process of low-performing DRs and retains good DRs to minimize unwanted mutations.

To ensure the legality of the DR, the head symbols of the gene can be replaced with any symbol from the FS and TS. The tail symbol of the

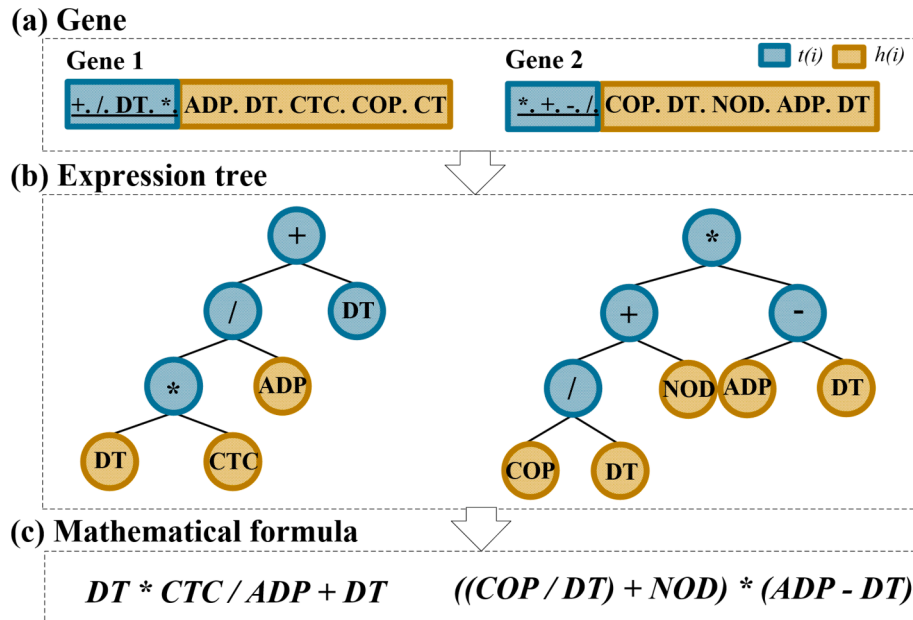


Fig. 5. Mapping between gene and DR in GEP.

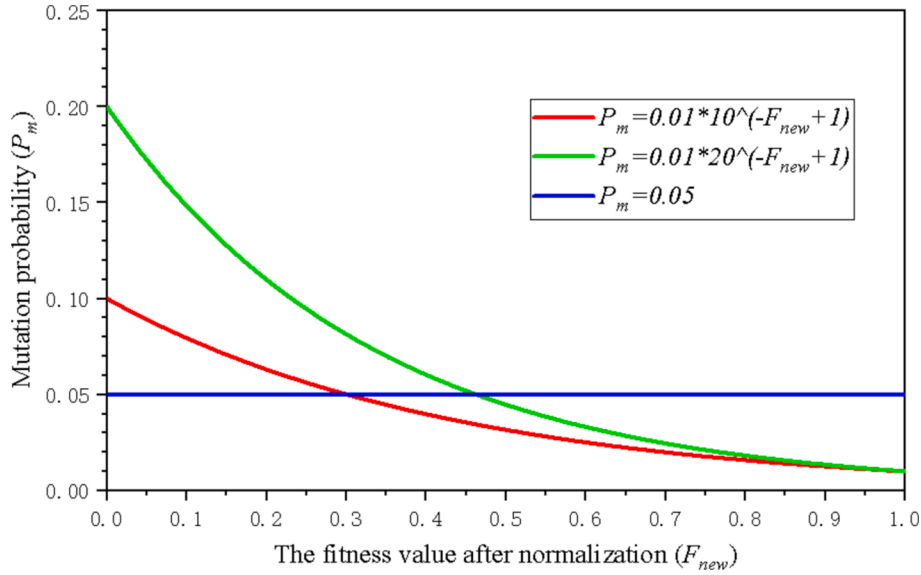


Fig. 6. Mutation probability mapping function.

gene can only be replaced with a symbol from the TS. Along with the mutation operator, the IGEP also utilizes a selection operator and a crossover operator.

Selection: The selection operator uses a roulette wheel method to draw out the retained individuals based on their similarity. Meanwhile, the optimal individuals are cloned and retained for the next generation.

Crossover: Multi-crossover is adopted. Multi-crossover points are first randomly selected. The elements on these points in the target solution are removed, and then the nonrepetitive elements of another solution fill these empty positions in turn.

6.3. Evaluation module

The evaluation module primarily consists of two processes: (1) transforming the newly generated DRs from the learning module into a feasible disassembly sequence; (2) calculating the product disassembly profit using the derived disassembly sequence and utilizing it as the fitness value of DRs to steer the evolution of the DRs in the learning module.

To translate the rules into a feasible disassembly sequence, the rule-driven disassembly sequence real-time planning model illustrated in Fig. 2 and Algorithm 1 is employed. However, unlike real-time data used in online planning, the data in the evaluation module becomes historical data. These historical data is transformed into matrix form, with the matrix information being gradually updated based on the rule decision. This simulates the dynamic information acquisition process in online planning. For each output DRs from the learning module, a feasible disassembly sequence is generated using historical data. Subsequently, the disassembly sequence generated by each DRs is used to compute the product disassembly profit using Eq. (15). Finally, the computed product disassembly profit is fed back to the learning module as the fitness value of the rule, facilitating the evolution of the rule population.

$$f = \sum_{j=1}^J \sum_{s=1}^S I_{sj} R_s^M x_j - C \sum_{j=1}^J T_j^M x_j - C \sum_{j=1}^J \sum_{k=1}^J \sum_{m=1}^{J-1} t_{jk} y_{jm} y_{k,m+1} \quad (15)$$

Eq. (15) closely resembles Eq. (1). However, unlike in Eq. (1), the subassembly recycling income is denoted as R_s^M representing the final recognition of subassembly recycling income after completing all disassembly operations. Similarly, T_j^M represents the final confirmed disassembly operation time after completing all disassembly operations.

7. Experiments and results

To investigate the performance of the proposed rule-driven method, a set of experiments is conducted. Two disassembly instances are used, i.e., the hammer drill, and the microwave oven.

7.1. Instance construction

To verify the generalization ability of the method across different product types, we conducted case studies using two products, a hammer drill and a microwave oven, which exhibit significant differences in structure, function, and components. This approach allowed for a more comprehensive evaluation of the algorithm's effectiveness (Fu et al., 2021). Specifically, the hammer drill set contains 26 subassemblies and 23 disassembly operations, and a microwave oven comprises 40 subassemblies and 34 disassembly operations. The exploded view of the hammer drill and microwave oven are shown in Fig. 7. The algorithm was tested using a variety of instance sizes, including 10, 20 and 30 instances, to determine the appropriate problem instance size. It was observed that 20 instances provide a sufficiently representative instance size for stable and reliable testing while avoiding the fluctuations in results that can occur with a smaller instance size. Consequently, 20 problem instances were selected as the final size. Furthermore, based on equipartition strategies in the literature (Li, Gao, Li, & Gao, 2020; Zhang et al., 2021), the 20 problem instances were divided into 2 groups, with 10 instances in each group designated as the training set and the testing set, respectively. Each instance of training set is used to evolve DRs. The best evolved DRs are then applied to the problems in the other group for validation. Each simulation experiment consisted of 20 different runs (or replications).

7.2. Comparison design

To examine the performance of proposed method, four types of algorithms, i.e., genetic programming (GP), GEP, IGEP^(0.01-0.1), and IGEP^(0.01-0.2) are chosen. Table 4 shows the parameter settings of the comparison algorithm. They are chosen because:

1) GP is a classic hyper-heuristic algorithm (Zhang, Mei, Nguyen, Tan, & Zhang, 2021). The GP is encoded using a tree structure where each node represents a function or termination symbol. The algorithm constructs the tree structure by combining operators and operands to represent the rule of the problem. GP is used to compare the

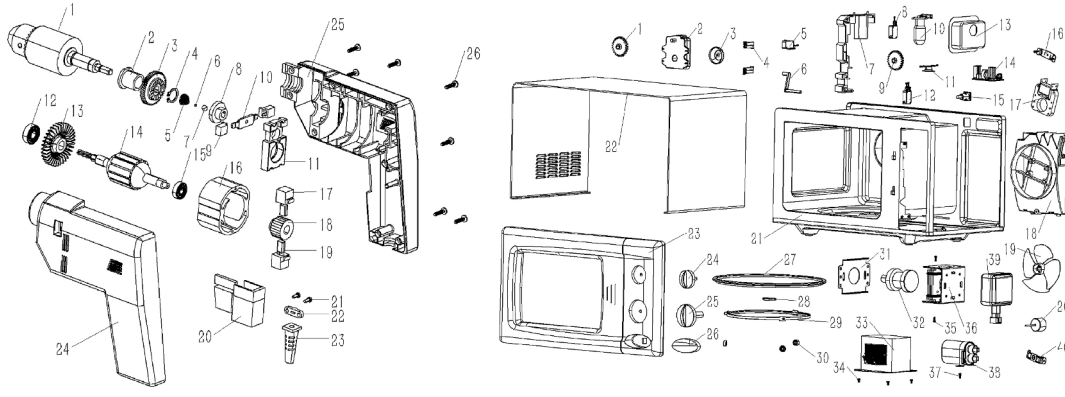


Fig. 7. Exploded view of the hammer drill and microwave oven.

Table 4
Parameter settings for GP, GEP, and IGEP.

Parameter	GP	GEP, IGEP ^(0.01-0.1) , and IGEP ^(0.01-0.2)
Termination condition	100 generations	100 generations
Population size	50	50
DR coding	The maximum depth does not exceed 6	The gene head length is 8
Selection strategy	Linear ranking selection	Linear ranking selection
Mutation probability	0.05	0.05, (0.01–0.1), and (0.01–0.2)
Crossover strategy	Subtree crossover	Two-points crossover
Crossover probability	0.2	0.2

performance of different hyper-heuristic algorithms in mining DRs;

2) GEP, IGEP^(0.01-0.1), and IGEP^(0.01-0.2) denote the GEP employing the three mutation probability mapping functions. The effect of the mutation probability mapping function is demonstrated by comparing the performance of three algorithms. Where, 0.01–0.1 indicates that the algorithm uses the mutation probability mapping function with the minimum value of the mutation probability of 0.01 and the maximum value of 0.1, i.e., $P_m = 0.01 \times 10^{(-F_{new}+1)}$.

7.3. Results and discussion

First the optimal disassembly rule mined by the IGEP^(0.01-0.1) on the training set are given and analyzed around the structure of the disassembly rules and the elements that compose them. Further, the performance of GP, GEP, IGEP^(0.01-0.1) and IGEP^(0.01-0.2) is compared in terms of efficiency and effectiveness in mining optimal disassembly rule: (2) In the training case, the training time, iterations, and convergence curves are compared to analyze the efficiency of algorithms in mining DRs; (3) In the testing case, the effectiveness of the optimal DR mined by the algorithms is analyzed by the compare of mean, standard deviation, and optimal of product disassembly profit.

(1) Disassembly rule analysis

Table 5 presents a set of solutions obtained by applying the optimal DR acquired through 20 runs of IGEP on the training set to a case in the test set. The table includes instance number (No), DR representing the disassembly rule mined from the training set, and the resulting disassembly sequence on a testing case. As an example, taking the first hammer drill solution shown in the table, the optimal DR mined from the training set is $CDP/NOD-DT \times C-ST$. By applying this rule to a test case in the test set, the resultant disassembly sequence is $1 \rightarrow 4 \rightarrow 9 \rightarrow 7 \rightarrow 22 \rightarrow 18 \rightarrow 30$, yielding a maximum product disassembly profit of 65.27 \$.

Analysis of the primary components of disassembly rules reveals that

Table 5
DRs and disassembly sequence found by IGEP^(0.01-0.1).

Product	No	DRs	Disassembly sequence	Profit (\$)
Hammer drill	1	$CDP-NOD-DT \times C-ST$	$1 \rightarrow 4 \rightarrow 9 \rightarrow 7 \rightarrow 22 \rightarrow 18 \rightarrow 30$	65.27
	2	$CDP-CT \times C-NOD$	$1 \rightarrow 4 \rightarrow 9 \rightarrow 16 \rightarrow 7 \rightarrow 22 \rightarrow 18 \rightarrow 29$	63.43
Microwave oven	1	$ROP-COP-CDP-CTC-NOD$	$2 \rightarrow 11 \rightarrow 12 \rightarrow 21 \rightarrow 22 \rightarrow 17 \rightarrow 29 \rightarrow 7$	89.67
	2	$ADP-COP+ROP-NOD$	$2 \rightarrow 11 \rightarrow 12 \rightarrow 22 \rightarrow 32$	82.34

CDP (Current disassembly profit), COP (The cost of performing the disassembly operating), and NOD (Number of disassembly operations done) play crucial roles in evaluating the priority of disassembly operations. Moreover, upon scrutinizing the structure of the rules, it can be found that the rule prefers to select the disassembly operation with larger recycling income of operating (ROP) and smaller cost of performing the disassembly operating (COP) as the next disassembly operation to be executed, which is also in line with what we expect in practice.

(2) Efficiency of algorithm

The hyper-heuristic algorithm constructs DRs from the problem instances in the training set.

Fig. 8 provides the convergence curves of the average product disassembly profit for GP, GEP, IGEP^(0.01-0.1), and IGEP^(0.01-0.2) over 20 independent runs on separate training sets for both the hammer drill and microwave oven. These convergence curves serve as a visual representation of the algorithm performance in mining DRs. The results shown in Fig. 8 demonstrate that GP and GEP exhibit remarkably similar convergence patterns, with almost identical iterations required for convergence. For example, in the case of the hammer drill, GP and GEP converge after 62 and 60 iterations. Similarly, for the microwave oven, GP and GEP reach convergence at 76 and 70 iterations. In contrast, IGEP^(0.01-0.1) and IGEP^(0.01-0.2) show significantly fewer convergence iterations compared to GP and GEP. This observation aligns with our expectations, as the fitness-driven mutation operator implemented in IGEP^(0.01-0.1) and IGEP^(0.01-0.2) aids rules with lower fitness in breaking through local optima, thus preventing premature convergence or getting trapped in suboptimal solutions. Simultaneously, rules with higher fitness are subjected to a lower mutation rate to preserve their advantageous gene combinations from unnecessary mutations. This fitness-driven mutation approach contributes to the improved convergence speed exhibited by IGEP^(0.01-0.1) and IGEP^(0.01-0.2).

To further compare the efficiency of the four algorithms in mining DRs, Fig. 9 and Fig. 10 give the specific values and box plots of the mean CPU running time of the four algorithms under 20 independent runs in 10 training instances. Table 6 presents the results of a Tukey HSD

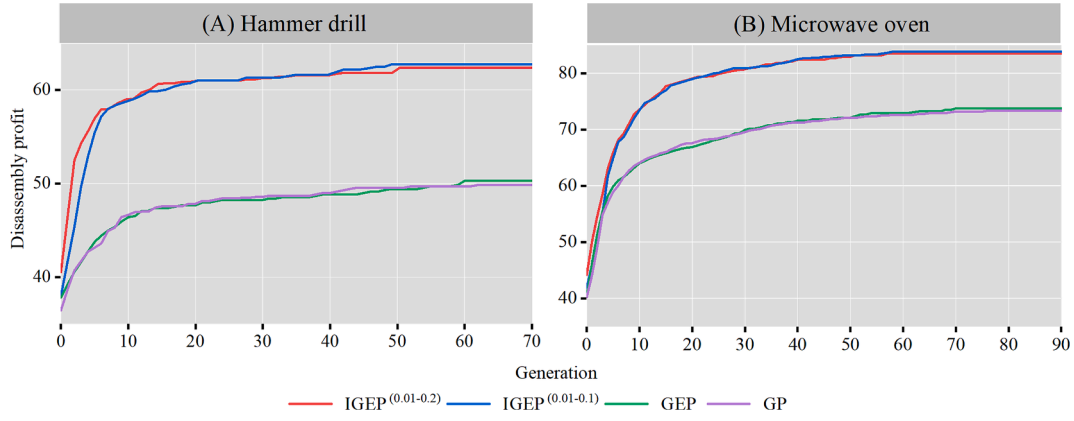


Fig. 8. The curve of average product disassembly profit under 20 independent runs of four algorithms.

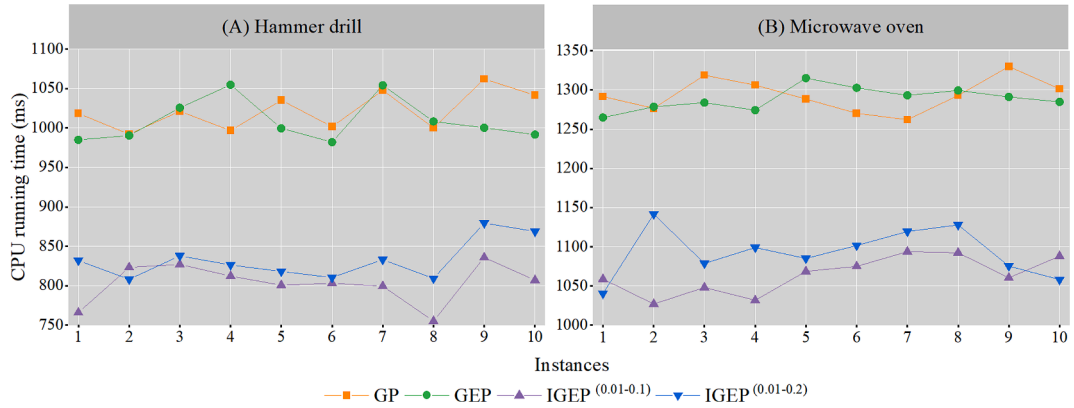


Fig. 9. Mean CPU running times according to 20 independent runs on 10 training instances of four algorithms in hammer drill and microwave oven.

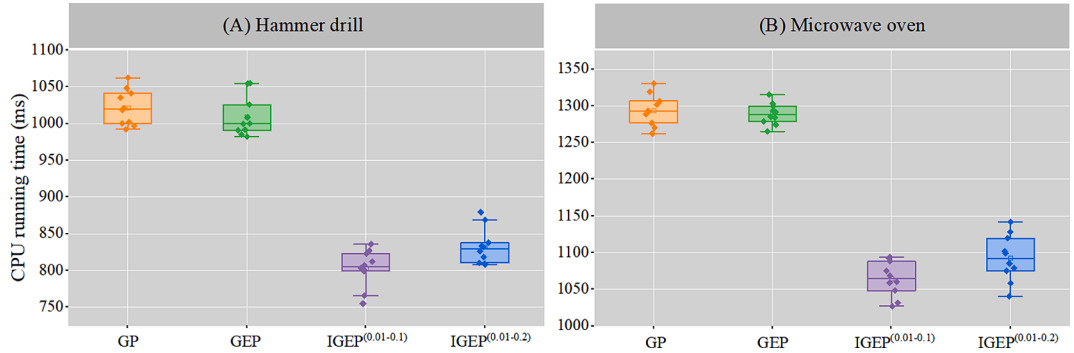


Fig. 10. Box plot of mean CPU running times on 10 training instances of four algorithms in hammer drill and microwave oven.

statistical test, analyzing the significance of differences in average CPU runtime among four algorithms across 10 training instances of hammer drill and microwave oven. The box plots in Fig. 10 show that the median CPU running times of GP and GEP are very close, and most of the running times fall within similar ranges. In addition, in Table 6 the P-values (0.5415 and 0.5323) of the difference in average CPU run time between GP and GEP are both greater than 0.05. This indicates that any observed differences in running times are likely due to random variation rather than systematic differences between the two algorithms. Therefore, GP and GEP have similar performance in DRs mining.

In contrast, IGEP^(0.01-0.1) generally exhibits shorter training times compared to IGEP^(0.01-0.2). For most training instances, such as hammers, IGEP^(0.01-0.1) demonstrates shorter training times in 9 out of 10 datasets, while microwaves show this trend in 8 out of 10 datasets.

Meanwhile, the significance analysis reveals p-value less than 0.05 between IGEP^(0.01-0.1) and IGEP^(0.01-0.2), indicating a significant difference between them. Thus, IGEP^(0.01-0.1) performs more efficiently in rule mining than IGEP^(0.01-0.2). A possible explanation for this observation is that the maximum mutation probability of 0.1 used by IGEP^(0.01-0.1) effectively enhances the algorithm optimization capabilities. In contrast, a higher mutation probability of 0.2 in IGEP^(0.01-0.2) introduces more instability, causing the algorithm to oscillate between multiple local optima, which delays convergence.

(3) Effectiveness of algorithm

Table 7 and Table 8 present the results of the four algorithms in terms of the mean, standard deviation, and optimal product disassembly profit value obtained for 20 independent runs in 20 testing cases of Hammer drill and Microwave oven. Among the four algorithms, IGEP^(0.01-0.1) and

Table 6

Significance analysis of average CPU runtime for four algorithms using Tukey HSD test.

Comparison	Hammer drill		Microwave oven	
	P-value	Significance	P-value	Significance
GP vs. GEP	0.5415	insignificant	0.5323	insignificant
GP vs. IGEP ^(0.01-0.1)	2.824E-13	$P < 0.05$	1.026E-14	$P < 0.05$
GP vs. IGEP ^(0.01-0.2)	6.045E-12	$P < 0.05$	2.015E-12	$P < 0.05$
GEP vs. IGEP ^(0.01-0.1)	1.803E-12	$P < 0.05$	1.584E-15	$P < 0.05$
GEP vs. IGEP ^(0.01-0.2)	4.120E-11	$P < 0.05$	7.007E-13	$P < 0.05$
IGEP ^(0.01-0.1) vs. IGEP ^(0.01-0.2)	0.0309	$P < 0.05$	0.0356	$P < 0.05$

IGEP^(0.01-0.2) achieved the highest mean product disassembly profit, the lowest standard deviation, and the optimal value across all 20 testing instances. Comparing IGEP^(0.01-0.1) and IGEP^(0.01-0.2), it is observed that IGEP^(0.01-0.1) outperformed IGEP^(0.01-0.2) in 15 out of 20 testing sets in terms of the mean value, and in 14 out of 20 testing sets in terms of the optimal value. Furthermore, in 13 out of 20 testing sets the standard deviation of IGEP^(0.01-0.1) is smaller than that of IGEP^(0.01-0.2). This indicates that IGEP^(0.01-0.1) more consistently found the optimal solutions compared to IGEP^(0.01-0.2) in hammer drill and microwave oven. This is mainly due to the larger mutation probability in IGEP^(0.01-0.2), which resulted in increased randomness and oscillation, hindering stable convergence towards better solutions. The lower mutation probability range (0.01–0.1) strikes a more effective balance between enhancing optimization capabilities and avoiding sub-optimal solutions that may result from high mutation rates.

Furthermore, a trend is observed when comparing the performance of IGEP^(0.01-0.1) and IGEP^(0.01-0.2) across two different products. As

product complexity increases, the performance gap between the two algorithms narrows. For instance, in the Hammer drill, IGEP^(0.01-0.1) outperforms IGEP^(0.01-0.2) in terms of average profit in 8 out of 10 instances. Conversely, in the Microwave oven tests, IGEP^(0.01-0.1) achieves higher average profit than IGEP^(0.01-0.2) in only 6 out of 10 instances. Additionally, the advantage of IGEP^(0.01-0.1) in standard deviation and optimal value also diminishes with increased of product complexity. This reduction is mainly due to the increased complexity of the microwave oven, which enlarges the scale and difficulty of the problem, thereby necessitating more flexible search strategies to find the optimal solution. The higher mutation probability of IGEP^(0.01-0.2) helps to improve the search ability of the algorithm in complex problems. Therefore, when using mutation probability mapping function to enhance algorithm performance, it is necessary to reasonably adjust the range of mutation probabilities according to the problem complexity to obtain the optimal algorithm performance.

8. Conclusion

The aim of this article is to develop a rule-driven disassembly sequences real-time planning method to solve the real-time planning of disassembly sequences of intelligent mixed-flow disassembly lines by mining disassembly rules. The goal is achieved by advancing progress in three key areas.

(1) A multi-matrix collaboration disassembly feature information representation matrix is designed to dynamically record the internal disassembly feature information obtained during the disassembly process. This approach effectively avoids the information bias associated with random, fuzzy data and probability representations, thereby enhancing the reliability of the planning process.

(2) A rule-driven real-time planning model is proposed to realize the real-time planning of selective disassembly sequences. This model prevents production chaos that may arise from pre-planning and frequent disturbance re-planning, ensuring the timeliness and efficiency of

Table 7

Mean, standard deviation, and optimal product disassembly profit value according to 20 independent runs on testing sets of four algorithms in hammer drill.

Hammer drill	GP		GEP		IGEP ^(0.01-0.1)		IGEP ^(0.01-0.2)	
	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal
1	57.62(4.40)	59.91	55.64(4.52)	56.41	63.21(3.71)	65.27	60.05(4.13)	63.71
2	50.78(4.57)	56.37	53.01(4.86)	57.05	61.17(3.54)	63.43	60.18(3.74)	62.16
3	56.45(4.76)	60.25	57.42(4.59)	60.39	67.85(3.64)	68.34	65.47(3.91)	67.10
4	47.61(4.75)	51.30	47.38(4.63)	52.14	55.87(3.76)	57.24	54.27(4.03)	56.67
5	51.92(5.01)	58.64	53.07(4.77)	56.41	66.29(3.23)	68.51	66.47(3.12)	69.22
6	52.02(4.62)	54.02	51.67(4.57)	54.37	57.13(3.51)	59.30	57.99(3.55)	58.93
7	54.17(4.52)	56.30	52.79(4.49)	55.22	65.15(3.48)	67.33	64.46(4.26)	66.43
8	50.32(4.78)	54.61	52.96(4.72)	56.06	60.16(3.42)	61.66	59.89(3.87)	62.26
9	57.53(4.84)	62.18	59.36(4.67)	64.72	69.78(3.13)	72.13	67.52(3.07)	70.04
10	50.68(4.81)	57.64	53.04(4.61)	58.20	64.31(3.60)	65.26	61.89(3.71)	63.24

Table 8

Mean, standard deviation, and optimal product disassembly profit value according to 20 independent runs on testing sets of four algorithms in microwave oven.

Microwave oven	GP		GEP		IGEP ^(0.01-0.1)		IGEP ^(0.01-0.2)	
	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal	Mean (standard deviation)	optimal
1	73.61(5.09)	77.38	75.38(5.18)	78.50	87.25(4.12)	89.67	85.56(3.74)	89.07
2	70.39(4.85)	72.90	70.72(5.03)	73.62	80.01(4.05)	82.34	75.57(4.26)	78.76
3	71.93(4.74)	75.50	72.04(4.75)	75.43	86.87(3.90)	88.35	88.23(3.78)	90.62
4	75.21(4.63)	77.95	73.61(4.89)	77.42	81.45(3.87)	83.62	80.71(4.05)	85.93
5	74.78(4.52)	76.85	72.99(4.31)	76.64	86.33(3.92)	88.18	82.67(4.24)	84.47
6	74.01(4.45)	77.03	73.27(5.02)	77.39	85.72(4.23)	87.72	87.21(4.16)	92.46
7	72.89(4.97)	77.32	71.15(4.87)	76.17	82.14(4.06)	86.78	83.24(3.91)	86.78
8	73.46(4.62)	76.68	73.78(4.81)	76.20	83.96(3.71)	86.14	81.24(3.88)	85.72
9	71.64(5.10)	74.16	71.85(5.12)	76.91	80.67(4.17)	84.31	79.43(4.13)	82.57
10	75.16(4.84)	79.15	74.32(4.95)	78.14	85.21(3.72)	86.14	82.44(4.17)	88.32

disassembly sequence planning.

(3) An IGEP based disassembly rule mining algorithm is developed to extract explicit disassembly rules from a large number of historical data sets generated during the disassembly process. The results show that the disassembly rules mined by hyper-heuristic algorithm are more likely to select disassembly operations with lower costs and higher recovery revenues, which is consistent with observations in practice. In addition, the mutation probability mapping function significantly improves the mining efficiency and rule quality of the GEP algorithm. Depending on the complexity of the product, it is important to adjust the range of mutation probabilities to achieve a balance between optimization capability and avoiding suboptimal solutions.

The explicit disassembly rules mined by the proposed method can help production managers in remanufacturing companies to make clearer and more rational decisions, rather than relying on implicit experience-based judgments. This approach effectively helps avoid losses due to errors in subjective decision-making. Meanwhile, the proposed mutation probability mapping function will be useful for improving the optimization ability of the hyper-heuristic algorithm.

Some interesting directions can be further investigated in the future. We plan to apply the proposed method to solve the integration problem of disassembly sequence planning and disassembly scheduling, and consider more optimization objectives such as energy consumption and safety considerations. In addition, we noted that the information in the disassembly feature matrices is calculated based on machine vision recognition results, which may introduce certain errors. The possible inaccurate information in the actual application process will have an impact on the performance of the method proposed in this paper. How to improve the quality of information in the matrix through data enhancement, multi-sensor data integration and other methods in practical application, to ensure the performance of the method will be the focus of our next research.

CRedit authorship contribution statement

Xin Chen: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Shuo Zhu:** Writing – review & editing, Methodology, Funding acquisition, Data curation. **Zhigang Jiang:** Supervision, Project administration. **Wei Yan:** Resources, Project administration. **Yanan Wang:** Resources. **Hua Zhang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research paper is supported by the National Natural Science Foundation of China (Grant No. 51905392, Grant No. 52375508) and “Chunhui Project Foundation of the Education Department of China (HZKY20220338). These financial contributions are gratefully acknowledged.

Data availability

Data will be made available on request.

References

- Behdad, S., Berg, L. P., Thurston, D., & Vance, J. (2014). Leveraging virtual reality experiences with mixed-integer nonlinear programming visualization of disassembly sequence planning under uncertainty. *Journal of Mechanical Design*, 136(4), Article 041005.
- Behdad, S., Williams, A. S., & Thurston, D. (2012). End-of-life decision making with uncertain product return quantity. *Journal of Mechanical Design - Transactions of the ASME*, 134(10), Article 100902.
- Branke, J., Nguyen, S., Pickardt, C. W., & Zhang, M. (2015). Automated design of production scheduling heuristics: A review. *IEEE Transactions on Evolutionary Computation*, 20(1), 110–124.
- Chand, M., & Ravi, C. (2023). A state-of-the-art literature survey on artificial intelligence techniques for disassembly sequence planning. *CIRP Journal of Manufacturing Science and Technology*, 41, 292–310.
- Drake, J. H., Kheiri, A., Özcan, E., & Burke, E. K. (2020). Recent advances in selection hyper-heuristics. *European Journal of Operational Research*, 285(2), 405–428.
- ElSayed, A., Kongar, E., Gupta, S. M., & Sobh, T. (2012). A robotic-driven disassembly sequence generator for end-of-life electronic products. *Journal of Intelligent & Robotic Systems*, 68, 43–52.
- Fu, Y., Zhou, M., Guo, X., Qi, L., & Sedraoui, K. (2021). Multiverse optimization algorithm for stochastic biobjective disassembly sequence planning subject to operation failures. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 52(2), 1041–1051.
- Gong, Q. S., Zhang, H., Jiang, Z. G., Wang, H., Wang, Y., & Hu, X. L. (2019). Nonempirical hybrid multi-attribute decision-making method for design for remanufacturing. *Advances in Manufacturing*, 7, 423–437.
- Gulivindala, A. K., Bahubalendruni, M. R., Varupala, S. P., & Ravi, C. (2021). Exponential moving average modelled particle swarm optimization algorithm for efficient disassembly sequence planning towards practical feasibility. *International Journal of Performance Engineering*, 17(3), 289.
- Güngör, A., & Gupta, S. M. (2001). Disassembly sequence plan generation using a branch-and-bound algorithm. *International Journal of Production Research*, 39(3), 481–509.
- Guo, X., Zhang, Z., Qi, L., Liu, S., Tang, Y., & Zhao, Z. (2022). Stochastic hybrid discrete grey wolf optimizer for multi-objective disassembly sequencing and line balancing planning in disassembling multiple products. *IEEE Transactions on Automation Science and Engineering*, 19, 1744–1756.
- Guo, X., Zhou, M., Abusorrah, A., Alsokhry, F., & Sedraoui, K. (2020). Disassembly sequence planning: A survey. *IEEE/CAA Journal of Automatica Sinica*, 8(7), 1308–1324.
- Kucukkoc, I., Keskin, G. A., & Li, Z. (2021). In 4 *A Differential Evolution Algorithm for Balancing Mixed-Model Disassembly Lines* (pp. 59–86). CRC Press.
- Laili, Y., Tao, F., Pham, D. T., Wang, Y., & Zhang, L. (2019). Robotic disassembly re-planning using a two-pointer detection strategy and a super-fast bees algorithm. *Robotics and Computer-Integrated Manufacturing*, 59, 130–142.
- Laili, Y., Ye, F., Wang, Y., & Zhang, L. (2021). Interference probability matrix for disassembly sequence planning under uncertain interference. *Journal of Manufacturing Systems*, 60, 214–225.
- Lee, S. C., Tseng, H. E., Chang, C. C., & Huang, Y. M. (2020). Applying interactive genetic algorithms to disassembly sequence planning. *International Journal of Precision Engineering and Manufacturing*, 21, 663–679.
- Li, X., Gao, L., Li, X., & Gao, L. (2020). GEP-based reactive scheduling policies for dynamic FJSP with job release dates. *Effective Methods for Integrated Process Planning and Scheduling*, 405–428.
- Lu, Q., Ren, Y., Jin, H., Meng, L., Li, L., Zhang, C., & Sutherland, J. W. (2020). A hybrid metaheuristic algorithm for a profit-oriented and energy-efficient disassembly sequencing problem. *Robotics and Computer-Integrated Manufacturing*, 61, Article 101828.
- Ma, Y. S., Jun, H. B., Kim, H. W., & Lee, D. H. (2011). Disassembly process planning algorithms for end-of-life product recovery and environmentally conscious disposal. *International journal of production research*, 49(23), 7007–7027.
- Mao, J., Hong, D., Chen, Z., Changhai, M., Weiwen, L., & Wang, J. (2021). Disassembly sequence planning of waste auto parts. *Journal of the Air & Waste Management Association*, 71(5), 607–619.
- Mircheski, I. (2019). Determination of disassembly interference matrix and improved nondestructive disassembly sequences for the product. *Journal of Production Engineering*, 30–34.
- Münker, S., & Schmitt, R. H. (2022). CAD-based and/or graph generation algorithms in (dis) assembly sequence planning of complex products. *Procedia CIRP*, 106, 144–149.
- Nie, L., Shao, X., Gao, L., & Li, W. (2010). Evolving scheduling rules with gene expression programming for dynamic single-machine scheduling problems. *The International Journal of Advanced Manufacturing Technology*, 50, 729–747.
- Ong, S. K., Chang, M. M. L., & Nee, A. Y. (2021). Product disassembly sequence planning: State-of-the-art, challenges, opportunities and future directions. *International Journal of Production Research*, 59(11), 3493–3508.
- Slama, I., Ben-Ammar, O., Dolgui, A., & Masmoudi, F. (2021). Genetic algorithm and Monte Carlo simulation for a stochastic capacitated disassembly lot-sizing problem under random lead times. *Computers & Industrial Engineering*, 159, Article 107468.
- Tian, G., Ren, Y., Feng, Y., Zhou, M., Zhang, H., & Tan, J. (2018). Modeling and planning for dual-objective selective disassembly using AND/OR graph and discrete artificial bee colony. *IEEE Transactions on Industrial Informatics*, 15(4), 2456–2468.
- Ullrich, C., & Buscher, U. (2013). Flexible disassembly planning considering product conditions. *International Journal of Production Research*, 51(20), 6209–6228.
- Xing, Y., Wu, D., & Qu, L. (2021). Parallel disassembly sequence planning using improved ant colony algorithm. *The International Journal of Advanced Manufacturing Technology*, 113, 2327–2342.
- Ye, F., Perrett, J., Zhang, L., Laili, Y., & Wang, Y. (2022). A self-evolving system for robotic disassembly sequence planning under uncertain interference conditions. *Robotics and Computer-Integrated Manufacturing*, 78, Article 102392.
- Yu, B., Wu, E., Chen, C., Yang, Y., Yao, B. Z., & Lin, Q. (2017). A general approach to optimize disassembly sequence planning based on disassembly network: A case study

- from automotive industry. *Advances in Production Engineering & Management*, 12(4), 305–320.
- Zeng, P., Liang, L., & Duan, Z. (2023). Ecological and environmental impacts of mineral exploitation in urban agglomerations. *Ecological Indicators*, 148, Article 110035.
- Zhang, F., Mei, Y., Nguyen, S., Tan, K. C., & Zhang, M. (2021). Multitask genetic programming-based generative hyper-heuristics: A case study in dynamic scheduling. *IEEE Transactions on Cybernetics*, 52(10), 10515–10528.
- Zhang, C., Zhou, Y., Peng, K., Li, X., Lian, K., & Zhang, S. (2021). Dynamic flexible job shop scheduling method based on improved gene expression programming. *Measurement and Control*, 54(7–8), 1136–1146.
- Zhao, A., Liu, P., Li, Y., Xie, Z., Hu, L., & Li, H. (2023). Real-time selection system of dispatching rules for the job shop scheduling problem. *Machines*.
- Zhou, Z., Liu, J., Pham, D. T., Xu, W., Ramirez, F. J., Ji, C., & Liu, Q. (2019). Disassembly sequence planning: Recent developments and future trends. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, 233(5), 1450–1471.